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Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts

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The selective conversion of polyethylene (PE), polypropylene (PP), and mixtures of these two polymers to form products with high volume demand is urgently needed because current methods suffer from low selectivity, produce large quantities of greenhouse gases, or rely on expensive, single-use catalysts. The isomerizing ethenolysis of unsaturated polyolefins could be an energetically and environmentally viable route to propylene and isobutylene; however, noble-metal homogeneous catalysts and an unsaturated polyolefin are currently required and the process has been limited to PE. We show that the simple combination of tungsten oxide on silica and sodium on gamma-alumina transforms PE, PP, or a mixture of the two, including postconsumer forms of these materials, to propylene or a mixture of propylene and isobutylene in greater than 90% yield at 320°C without the need for dehydrogenation of the starting polyolefins.

The accumulation of plastic waste in landfills and the environment is a major economic and environmental issue that will intensify in the coming decades (*1*). Aliphatic polyolefins, almost entirely composed of polyethylene (PE) and polypropylene (PP), account for the majority (57%) of all polymer resins produced (*2*). These aliphatic polyolefins contain only C–H and C–C bonds, which are inert to almost all chemical transformations. Accordingly, aliphatic polyolefins are among the longest-lived waste plastics in the environment (*3*) and the least amenable to valorization by chemical modification.

Thermal pyrolysis of polyethylene and polypropylene at ≥500°C or catalytic thermolysis can convert these polyolefins to a complex mixture of hydrocarbons (Fig. 1A) (*4*, *5*) with varying levels of selectivity for olefin (*6*) or arene (*7*) products, and many systems for chain cleavage with hydrogen (hydrogenolysis) catalyzed by transition metals have been reported to convert PE and PP to mixtures of linear and branched alkanes (*8*–*13*). Tandem reactions also have been reported to cleave polyolefins. Net alkane metathesis that combines PE with an excess of light alkanes yields mixtures of short alkanes by tandem alkane dehydrogenation and olefin metathesis (*14*–*16*), and tandem dehydrogenation, aromatization, and hydrogenolysis converts polyethylene into mixtures of alkylarenes (*17*, *18*). Pyrolysis and hydrogenolysis often produce substantial quantities of methane that contribute to greenhouse gas emissions, whether they are released, used as fuel, or steam-reformed. Each of these methods produces distributions of hydrocarbons that necessitate difficult separations or further downstream reactions (*19*). Further, the products from the pyrolysis of waste polyolefins are highly sensitive to the composition of the waste input, reactor design, and heat transfer, and the presence of oxygenated impurities in the waste feedstock often necessitates downstream hydrogenation to eliminate unstable oxygenated products (*19*). Thus, methods are required that generate easily separable products with high selectivity, while producing little or no methane. In addition, the massive scale of polyolefin waste (>160 megatonnes in 2015) makes it necessary to generate products for which there is a similarly large demand (*2*).

Light olefins, such as ethylene, propylene, and butenes, are among the largest volume commodity chemicals (*20*). Recently, our group and Scott, Guironnet, and co-workers (*21*, *22*) independently showed that isomerizing ethenolysis (IE)—a catalytic process that converts internal or terminal olefins to shorter terminal olefins by a combination of olefin isomerization and olefin metathesis (*23*, *24*)—can produce propylene with high selectivity from polyethylene that is dehydrogenated or that is formed with unsaturation at the chain end (*25*), as well as medium chain olefins (C_6-C_{18}) (*26*, *27*). However, this reaction has not been observed with PP, the second most abundant waste plastic. The IE of PP generates a mixture of propylene and isobutylene. Although the formation of two alkenes necessitates downstream separation, the purification of ethylene and propylene from a mixture of light olefins and paraffins by cryogenic distillation has been practiced for decades after steam and fluid catalytic cracking, among other processes (*28*–*32*). IE also does not generate ethane or propane and thereby avoids the most difficult separations in the conventional production of light olefins, namely the separation of ethylene from ethane and propylene from propane. The purification of ethylene, propylene, and isobutylene from the CIE of PP is therefore much simpler and less energy intensive than the purifications associated with the conventional production of light olefins.

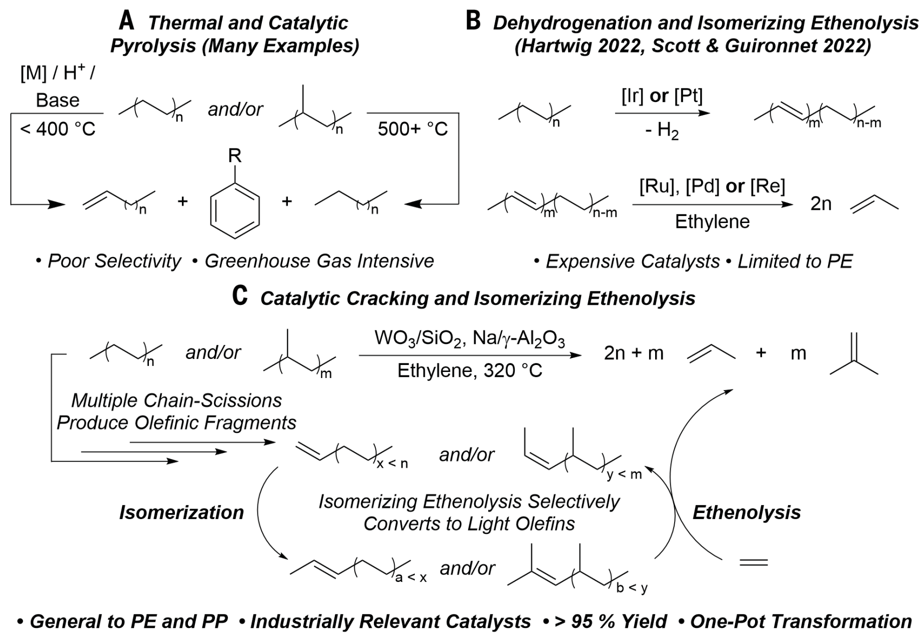

Fig. 1. Comparison with prior art. (**A**) Previous work on the application of thermal and catalytic cracking for the chemical recycling of polyolefins. (**B**) Previous work applying a strategy of dehydrogenation and tandem isomerizing ethenolysis to affect the conversion of PE to propylene. (**C**) Current work on the combination of catalytic cracking and isomerizing ethenolysis to yield light olefins from PE and PP.

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Because PE and PP are among the most difficult and expensive plastics to separate from each other in a mixed-waste stream, it is crucial that an IE process apply to both polyolefins. In addition, the previous IE reaction on PE required homogeneous catalysts that were expensive and not recyclable (*21*, *22*). These drawbacks create an urgent need to discover a base-metal heterogeneous catalyst system for the IE of both PE and PP to produce light olefins selectively.

We report a distinct approach to the deconstruction of both PE and PP to propylene and isobutylene that bypasses the need for polymer dehydrogenation or unsaturated polymer and occurs with a combination of base-metal catalysts comprising tungsten oxide on silica (WO_3/SiO_2) and sodium metal on gamma-phase alumina ($Na/\gamma\text{-}Al_2O_3$). These reactions are initiated by a combination of catalytic cracking by the W- or Na-containing catalysts, which then convert PE and PP to propylene and a combination of propylene and isobutylene, respectively, with >90% yield in 90 min on a 1-g scale without competing generation of methane. The reaction occurs at 320°C (close to that used for catalytic cracking) because the reaction appears to initiate by catalytic cleavage of the C–C bonds of the alkane chains to generate alkenes. The combination of a basic olefin isomerization catalyst and the electrophilic WO_3/SiO_2 as olefin metathesis catalyst makes this system closely related to that of the industrial neohexene process, in which isomerizing ethenolysis of 2,4,4-trimethyl-1-pentene (diisobutylene) occurs with the combination of magnesium oxide and WO_3/SiO_2 as the catalysts (*33*). Because WO_3/SiO_2 has been used industrially as an olefin metathesis catalyst in olefin-conversion technology (>5 megatonnes/year) and in the neohexene process (> 250 kilotonnes/year) (*31*, *33*, *34*), and alkali metals supported on alumina have been used industrially as olefin isomerization catalysts for the Shell higher olefin process (>1 megatonnes/yr) (*31*, *35*), the conversion of PE to propylene and PP to propylene and isobutylene with these two catalysts alone constitutes a major step toward scalable, robust, and selective conversion of PE and PP and mixtures of the two to light olefins.

Catalytic cracking of polyolefins with $Na/\gamma\text{-}Al_2O_3$

We began our studies on polyolefin deconstruction to light olefins with heterogeneous catalysts by addressing the necessity of an initial dehydrogenation step to create unsaturation in the PE chains for IE. Prior work has shown that PE dehydrogenation can be achieved with a transition metal–based homogeneous catalyst, together with stoichiometric hydrogen acceptors, or with a transition metal–based heterogeneous catalyst. In the latter case, dehydrogenation is accompanied by aromatization and hydrogenolysis of the polymer. Hence, there is a need to identify heterogeneous catalysts for PE dehydrogenation that do not concurrently generate methane or arenes (*36*). Transfer dehydrogenation of PP with molecular catalysts has been demonstrated (*37*) but heterogeneous dehydrogenation of PP has not been reported.

We hypothesized that the difficulties confronting PE and PP dehydrogenation could be circumvented and the dependence on precious metals could be eliminated by using catalytic chain scission to form shorter polymer chains containing C=C bonds that could be used for subsequent IE. We further hypothesized that a basic catalyst could affect this chain cleavage step without promoting skeletal isomerization of PE and could catalyze isomerization of the olefins during concomitant IE (*38*). If so, then PE could be converted selectively to propylene, and PP could be converted selectively to propylene and isobutylene by a combination of catalytic cracking and isomerizing ethenolysis (CIE), as shown in Fig. 1C.

To investigate the feasibility of this strategy, we subjected isotactic PP [(iPP) Aldrich, M_n = 28.0 kilodaltons (kDa), 1.00 g scale] and high-density polyethylene (HDPE) (Aldrich, M_n = 9.73 kDa) to $Na/\gamma\text{-}Al_2O_3$ under 15 bar of ethylene at 320°C. After 90 min of heating, the iPP subjected to these conditions contained approximately 2.6 ± 0.6% trisubstituted olefin and 0.35 ± 0.07% 1,1-disubstituted olefin, with respect to propylene repeat units, as determined by proton nuclear magnetic resonance (^{1}H NMR) spectroscopy (Fig. 2A). Likewise, HDPE subjected to these conditions contained approximately 0.48 ± 0.11% internal olefin (SD, average of three runs). HDPE heated under the same conditions in the absence of catalyst contained 0.11 ± 0.02% (SD, average of three runs) terminal olefin and <0.01% internal olefin (Fig. 2A).

Analysis by high temperature size exclusion chromatography (HT-SEC) revealed that each of the recovered polymers underwent measurable reductions in molecular weight and that the reduction was similar for each sample. The number-average molecular weight (M_n) of the starting iPP was 28.0 kDa, whereas the M_n of the same iPP heated at 320°C in the presence of $Na/\gamma\text{-}Al_2O_3$ for 90 min was 1.63 kDa and that of the iPP heated at 320°C with ethylene alone for the same time was 1.75 kDa (Fig. 2B). For PE, the starting M_n was 9.73 kDa, whereas the M_n of the PE after heating at 320°C in the presence of $Na/\gamma\text{-}Al_2O_3$ for 90 min was 2.25 ± 0.15 kDa, and the M_n of the PE heated at 320°C for the same time in the absence of any additive was 3.18 ± 0.49 kDa. Thus, we conclude that heating PE and PP mediates scission of the polymer chains but heating in contact with $Na/\gamma\text{-}Al_2O_3$ can enhance the formation of olefins, along with this chain scission, particularly for PE.

Cracking and olefin metathesis of polyolefins with WO_3/SiO_2

Having seen that the cleavage of PE and PP to long-chain alkenes occurs at 320°C in the presence of $Na/\gamma\text{-}Al_2O_3$ and knowing from prior work that $Na/\gamma\text{-}Al_2O_3$ catalyzes alkene isomerization (*39*), we sought a similarly simple and robust catalyst for olefin metathesis to pair with the isomerization catalyst and tested whether

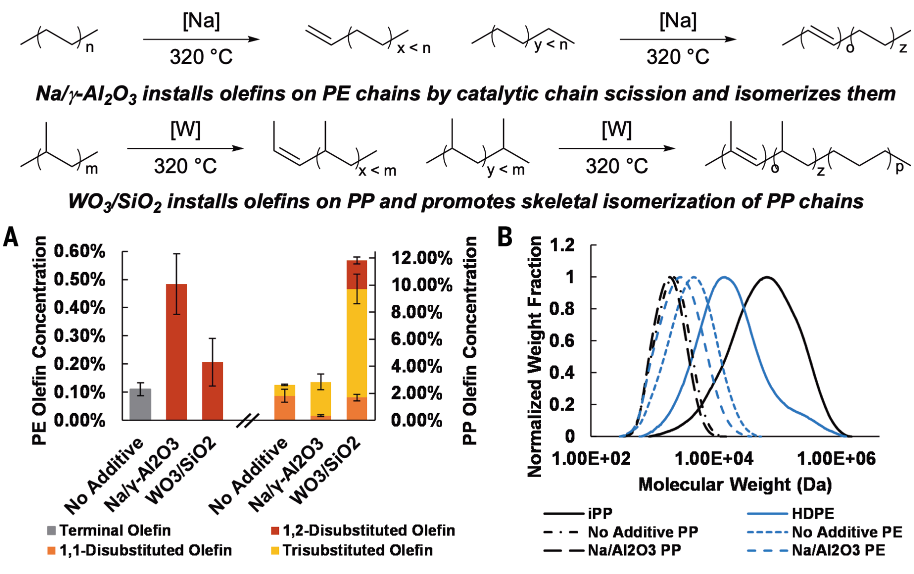

Fig. 2. Experiments on the catalytic cracking of polyolefins. (**A**) Amount of olefin in PE and PP chains after chain cleavage at 320°C in the presence of ethylene (10 bar) with no additive, $Na/\gamma\text{-}Al_2O_3$ catalyst, or WO_3/SiO_2 catalyst, determined by ^{1}H NMR spectroscopy. Reported values for olefin content are the average of three runs, with error bars representing the SD of the three runs. (**B**) Normalized distributions of the molecular weights of PE and PP chains after chain cleavage at 320°C in the presence of ethylene (10 bar) with and without $Na/\gamma\text{-}Al_2O_3$, determined by HT-SEC.

the metathesis catalyst could likewise catalyze cracking of the polyolefin chains.

We tested WO_3/SiO_2 as the olefin metathesis catalyst under the conditions for chain scission and isomerization because it has been reported to catalyze ethenolysis of both linear and branched olefins at temperatures between 300° and 400°C (*40–42*). Thus, we synthesized WO_3/SiO_2 according to previous work by Howell *et al.*, which showed that a catalyst with 0.6 W atoms per square nanometer on silica support was most active for the self-metathesis of propylene (*43*). Upon contacting iPP with WO_3/SiO_2 at 320°C under ethylene in the absence of Na/γ-Al_2O_3, we observed degradation of the polymer into an oil containing 11.8% olefin with respect to PP repeat units. This level of olefin formation is three times that observed in samples of iPP contacted with ethylene and Na/γ-Al_2O_3 (Fig. 2A). HDPE contacted with ethylene and WO_3/SiO_2 under the same conditions produced only 0.21 ± 0.09% internal olefin and possessed an M_n of 2.82 ± 0.44 kDa. These data show that iPP undergoes chain cleavage and that it forms olefins in the presence of the WO_3/SiO_2 to a much larger extent than does PE. They also show that iPP undergoes chain cleavage and formation of olefins to a larger extent in the presence of the WO_3/SiO_2 than it does in the presence of Na/γ-Al_2O_3, but that PE undergoes chain cleavage and formation of olefins in the presence of WO_3/SiO_2 to a smaller extent than it does in the presence of Na/γ-Al_2O_3.

CIE of polypropylene, polyethylene, and mixtures of polypropylene and polyethylene

Simple CIE of PP would yield one equivalent of propylene and one equivalent of isobutylene per monomer unit converted. However, the combination of WO_3/SiO_2 and Na/γ-Al_2O_3 under 15 bar of ethylene at 320°C converted various sources of PP to an approximately 2:1 ratio of propylene to isobutylene. For example, a commercial sample of iPP (Sigma-Aldrich, 28 kDa) converted to a mixture of 1.45 mol of propylene and 0.87 mol of isobutylene per mole of PP monomer units after 90 min (Fig. 3A). This combination of values represents quantitative conversion of the polypropylene chains to light olefins. The reaction of atactic polypropylene occurred with a slightly lower, but still high, yield of 1.37 mol of propylene and 0.74 mol of isobutylene per mole of PP monomer units, or 95% conversion of the full polypropylene chains to light olefins. The amount of propylene formed in this process is greater than the amount of isobutylene and greater than the amount the stoichiometry predicts. Because the ratio of propylene to isobutylene is proportional to the ratio of methylene units to methine units in the polypropylene backbone, we hypothesized that this 2:1 ratio of propylene to isobutylene could result from skeletal isomerization of the polypropylene backbone by the acidic WO_3/SiO_2 catalyst to convert a –CH(Me)– unit into a –CH_2CH_2– unit.

Consistent with this explanation for the stoichiometry of products, the sample of iPP contacted with WO_3/SiO_2 at 320°C for 90 min incorporated approximately 25.2% of the methyl groups into the polymer backbone as methylene units, as determined by ^{1}H NMR spectroscopy. This extent of isomerization corresponds to a theoretical yield of 1.50 mol of propylene and 0.75 mol of isobutylene per mole of PP monomer units, a ratio matching closely with the yields observed in our CIE experiments. The same isomerization was not observed for iPP contacted with Na/γ-Al_2O_3 under the same conditions. We envision that further enhancement of backbone isomerization could drive the selectivity of the CIE of PP toward propylene, enabling the full conversion of PP to the monomer from which it was made, an outcome that might be preferable if propylene were judged to be the preferred product because of the circularity created by reforming propylene, rather than a combination of propylene and isobutylene, which might be more desirable owing to the higher cost of isobutylene.

The combination of WO_3/SiO_2 and Na/γ-Al_2O_3 also catalyzed the deconstruction of various classes of polyethene selectively to propylene. The reaction of HDPE (Sigma-Aldrich, M_n = 9.10 kDa) with ethylene (15 bar) formed propylene in 87% yield [565 turnover number (TON) in tungsten (W), Fig. 3B] in less than 90 min (average turnover rate of 0.11 s^{-1}) at 320°C. More than 95% conversion of the polymer was observed. Of the remaining 5% of nonvolatile

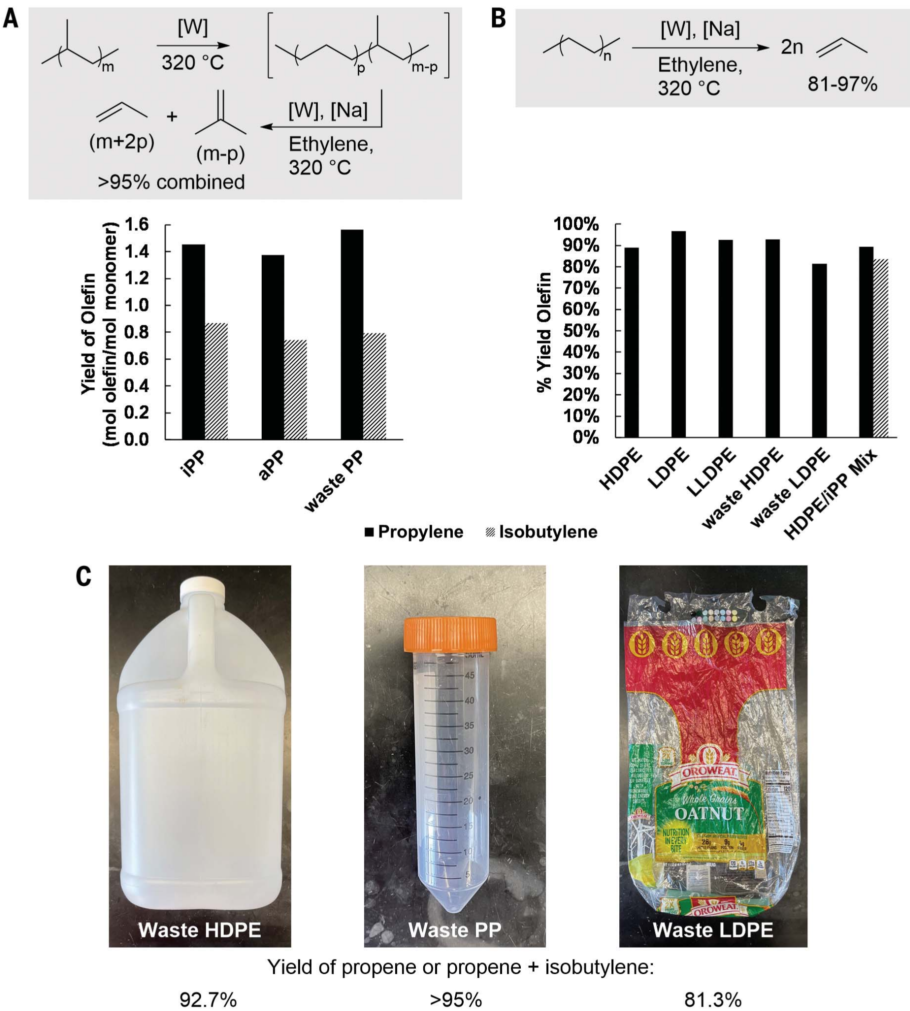

Fig. 3. Experiments on the tandem catalytic cracking and isomerizing ethenolysis of polyolefins (CIE). (**A**) Yields of CIE on various examples of commercial and waste polypropylene: PP (1.00 g, 23.8 mmol of monomer units), WO_3/SiO_2 (400.0 mg, 0.1075 mmol W), and Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) heated to 320°C under 15.0 bar of ethylene (175 mmol) for 90 min. (**B**) Yields of CIE on various examples of commercial and waste polyethylene: PE (1.00 g, 35.7 mmol of monomer units), WO_3/SiO_2 (400.0 mg, 0.1075 mmol W), and Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) heated to 320°C under 15.0 bar of ethylene (175 mmol) for 90 min. (**C**) Photographs of sources of waste subjected to CIE: waste HDPE (left), waste PP (middle), and waste LDPE from a bread bag (right).

components, three-fifths was extracted into dichloromethane and was isolated as a heavy oil consisting of C_6-C_{18} olefins. Concomitant consumption of ethylene was also observed; 62.5 mmol of ethylene were consumed and 62.2 mmol of propylene were produced, illustrating good closure of the mass balance for this process. In a similar fashion, commercial samples of low density polyethylene (LDPE) and linear low-density polyethylene (LLDPE) formed propylene with the same catalysts, temperature, and time in yields of 97 and 93%, respectively. The CIE of HDPE also occurred at a lower temperature; a similar yield of propylene (88%) was obtained after 18 hours at 280°C.

For each of these reactions, methane was used as the internal standard. To assess whether methane was generated during the reaction, we conducted an identical CIE of HDPE in the absence of added methane as an internal standard. Only 0.018 mmol of methane was generated in the reaction (assuming the same yield of propylene), amounting to <0.25% yield of methane from the HDPE.

The inability to separate PE from PP in a mixed-waste stream makes it important to demonstrate the deconstruction of a mixture of both polymers. Thus, we tested CIE on a 1.2:1 (w/w) mixture of HDPE to iPP. This reaction converted all of the polymer to a mixture of propylene and isobutylene in a 5.7:1 ratio, for yields of 89 and 83% propylene and isobutylene, respectively (see Fig. 3B). The ratio of propylene to isobutylene is slightly higher than expected for a 1.2:1 (w/w) mixture of PE to PP (4.6:1), and this high ratio is further consistent with backbone isomerization of the PP chains within the mixture of PE and PP.

CIE of postconsumer waste

We also tested the CIE of postconsumer waste catalyzed by this system (Fig. 3C). All examples of waste plastic were first processed by precipitation from boiling toluene into methanol and mechanical grinding to a fine powder. A 1.0-gallon jug selected as a source of waste HDPE produced propylene in a yield of 93% that is comparable to that for CIE of a commercial sample of virgin HDPE. A centrifuge tube selected as a source of waste polypropylene formed a 2.0:1 ratio of propylene and isobutylene in yields of 1.56 and 0.79 mol per mole of PP monomer units, respectively (which again corresponds to quantitative conversion of the polypropylene chains to light olefins), which are comparable to those for CIE of commercial, virgin iPP and again reflect some skeletal isomerization of the PP backbone. A bread bag was chosen as representative food packaging because it is a composite material containing several different polymers; this material also reacted in high yield. Characterization of this bread bag by quantitative ^1H NMR spectroscopy revealed that the material was 65% PE by mass. This source of LDPE, thus, yielded propylene in a yield of 81.3%, which is only slightly lower than that obtained from CIE of commercial, virgin LDPE.

To investigate the robustness of the CIE approach for use on a postconsumer waste stream, we conducted CIE on HDPE with added contaminants. We found that adding 5% (w/w) of PS to the reaction mixture did not affect the CIE of HDPE to a large extent. However, addition of 5 wt% of polyvinyl chloride (PVC) or polyethylene terephthalate (PET) to the reaction mixture decreased the yield of propylene to 20.5 and 16.9%, respectively. Likewise, CIE of HDPE in the presence of 5.0 wt% of the plasticizer bis(2-ethylhexyl) phthalate resulted in a considerably diminished 25.7% yield of propylene. We note that the separation of polyolefins from contaminants, such as PET or PVC, by flotation is well-precedented, as is the removal of PVC or of catalyst-poisoning chlorine from a mixed-waste stream by thermal decomposition and recovery of the resulting HCl (*44*). Therefore, application of the CIE approach to a mixed-waste stream would require separation of the contaminants from the polymer prior to its deconstruction.

To assess whether the light olefins observed in the above experiments were produced from the polyolefin reactants and ethylene or from ethylene alone, we conducted an isotopic labeling experiment. We subjected a mixture of 95.5 wt% HDPE and 4.5 wt% $^{13}C_2$-PE and a mixture of 95.5 wt% iPP and 4.5 wt% ^{13}C-1 PP to our CIE approach. The resulting gaseous products of each CIE reaction were then analyzed by gas chromatography-mass spectrometry to determine the isotopic abundance of ^{13}C in the propylene product. We found a ^{13}C isotopic abundance of 8.35 ± 0.63% in the propylene originating from PE, and an abundance of 5.68 ± 1.46% in the propylene originating from PP (errors represent 1 SD resulting from three separate measurements) (Fig. 4A). These levels of ^{13}C enrichment correspond to 1.01 mol of ^{13}C propylene per monomer of the partially labeled PE and 0.84 mol of ^{13}C propylene per monomer of the partially labeled PP. These levels of enrichment also correspond to 0.89 mol of ^{12}C-propylene per monomer from the same PE and 1.30 mol of propene from the same sample of PP. As noted above, this yield of propylene

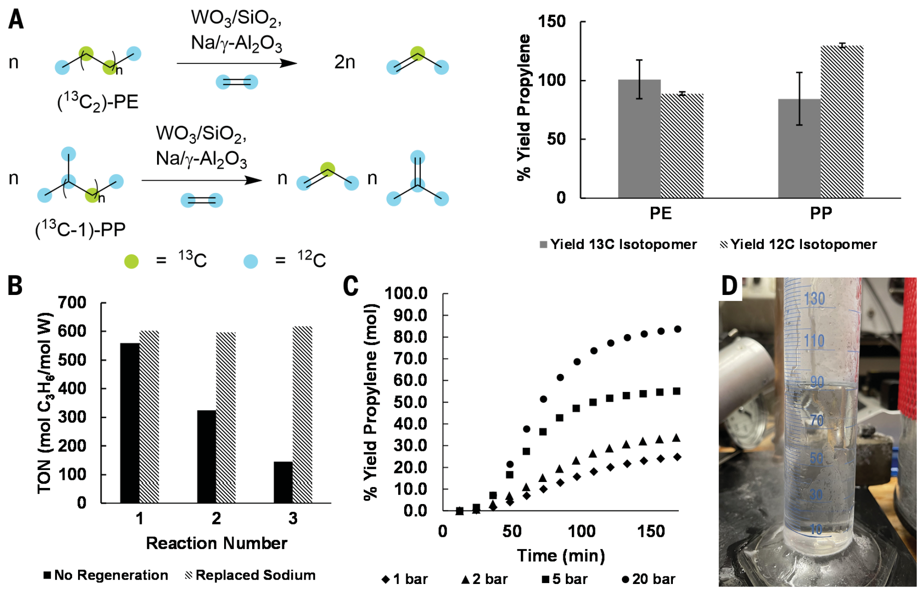

Fig. 4. Mechanism, catalyst reuse, and scale-up. (**A**) Isotopic labeling experiments with unlabeled PE and PP (^{13}C-1): unlabeled PE (954.8 mg, 34.04 mmol of monomer units) and $^{13}C_2$-PE (46.7 mg, 1.55 mmol) or unlabeled PP (958.5 mg, 22.78 mmol of monomer units) and ^{13}C-1 PP (45.8 mg, 1.06 mmol), WO_3/SiO_2 (400.0 mg, 0.1075 mmol W), and Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) were heated to 320°C under 15.0 bar of ethylene (175 mmol) for 90 min. The headspace of each reaction was then analyzed by GC-FID and GC-MS to determine overall propylene yields and enrichment in ^{13}C, respectively (graph, right). Reported yields of each isotopomer are the average of three runs, with error bars representing the SD of the three runs. (**B**) Investigation of catalyst recyclability: PE (1.00 g, 35.7 mmol of monomer units), WO_3/SiO_2 (400.0 mg, 0.1075 mmol W), and Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) were heated to 320°C under 15.0 bar of ethylene (175 mmol) for 90 min. The reactor was then dismantled under inert atmosphere, and additional PE (1.00 g, 35.7 mmol of monomer units) was added for two additional cycles (solid bars) or additional PE (1.00 g, 35.7 mmol of monomer units) and Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) were added in the second cycle, with only PE added in the third cycle (dashed bars). (**C**) Investigation of the effect of ethylene pressure on the yield of HDPE CIE in a semi-batch reactor setup. The legend denotes the pressure of ethylene in the reactor, and t = 0 was defined as when heating was initiated. In all cases, the reaction temperature was reached around t = 50 min. (**D**) Isolated yield of a 3.6:1 mixture of liquefied propylene and butenes from the CIE of 50 g of HDPE condensed at −94°C.

from PP results from parallel skeletal isomerization of the PP, thereby enriching the product mixture in propylene over ^{12}C-isobutylene. These high yields of ^{13}C-1 propylene eliminate the possibility of propylene in the above experiments being generated by tandem ethylene oligomerization and isomerizing ethenolysis of the ethylene oligomers. The yield of propylene from the CIE of PE in the above experiments with pure polymer, therefore, originated almost exclusively from the PE rather than from ethylene alone.

To determine the stability of this catalyst system, we conducted multiple cycles of CIE using the same batch of catalyst. After the CIE of HDPE was conducted as described above, we terminated the reaction and dismantled the reactor under inert atmosphere. An additional portion of HDPE was added to the same catalyst, and the reaction was conducted in an identical manner. This process was repeated three times. We observed that the system retained 50% activity after each subsequent run, for a total TON of 1030 with respect to W over the three runs (Fig. 4B). This value is more than an order of magnitude higher than the TON we reported previously for the IE of unsaturated HDPE using a molecular olefin metathesis catalyst and in this case occurs directly on HDPE without initial dehydrogenation (*21*). Moreover, addition of a fresh portion of the Na/γ-Al_2O_3 catalyst in the second reaction cycle completely restored the activity of the system for two new reaction cycles. This result indicates that the activity of WO_3/SiO_2 is nearly constant over three reaction cycles, but the activity of the Na/γ-Al_2O_3 catalyst decreases by approximately 50% with each subsequent reaction cycle.

Reactions on larger scale in a semi-batch reactor

Finally, we investigated the feasibility of conducting the CIE of polyolefins in a semi-batch reactor configuration. The CIE of HDPE under an ethylene flow of 10 standard cubic centimeters per minute (SCCM) at ambient pressure led to 26.9% yield of propylene, and reaction with an ethylene pressure of 20 bar and an ethylene flow of 200 SCCM led to a yield of propylene that was comparable to that of our batch experiments (84.5%, Fig. 4C). The CIE of PP under the same conditions yielded 1.54 mol of propylene and 0.79 mol of isobutylene per mole of propylene monomer unit. These values represent >95% conversion of the carbon in the PP monomer units to propylene and isobutylene. The selectivity for isobutylene over other butenes was 97%. Given that reactive separation of isobutylene from a mixture of butenes is one of the most substantial contributors to the cost of isobutylene production, this high selectivity presents an advantage of our CIE of PP over reactive separation for the production of this light olefin (*45*).

The alkenes formed with a continuous flow of ethylene, a decreased mol % of catalyst, and an increased amount of PE (50 g) were isolated by stepwise condensation of the efflux at 20° and −94°C after 3 hours of reaction. The fraction collected at −94°C constituted 90 mL (55 g if the density is that of liquid propylene at the boiling point) of a 3.6:1 mixture of liquefied propylene and butenes, representing 438 TON with respect to WO_3/SiO_2 during this run (Fig. 4D). The ability to obtain high yields from this reactor configuration demonstrates the potential of operating the CIE process on a larger scale.

Because of the considerably shorter residence time under these conditions, we also collected a large quantity of higher olefins. The 20°C fraction contained 20.0 g of a mixture of C_5-C_{16} olefins. Most of these higher olefins were pentenes (50.2%), hexenes (23.8%), and heptenes (13.2%), and the concentration of each olefin in the mixture was proportional to the vapor pressure of that olefin at 20°C (see fig. S19). Therefore, we conclude that higher olefins could be produced with high selectivity using the CIE method by careful engineering of reactor residence time and outlet temperature. At this larger scale, we found mass transport to strongly influence the reaction outcome. For this reason, we modified the impeller used to stir the reaction mixture to improve mass transport (fig. S16). The isolation of both light olefins and higher olefins from the reaction mixture demonstrates the flexibility of CIE to generate light olefins with high selectivity for either one or two products or to generate mixtures of higher olefins.

The results of this work demonstrate the feasibility of deconstructing PE and PP, the two largest-volume plastics, individually or in a mixture to form products that are feedstock materials for the chemical industry using inexpensive heterogeneous catalysts that can be recycled and used in a semi-batch reactor. Although a robust techno-economic assessment is required to establish the economic feasibility of conducting this reaction on commercial scales, we hope that the work described in this study will lead to practical methods for recovering the carbon in PE and PP as propylene and isobutylene, which can be used to produce new polymers. By doing so, the demand for production of these essential commodity chemicals starting from fossil carbon sources (e.g., petroleum and natural gas) and the associated greenhouse gas emissions could be greatly reduced.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S27
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Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts

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Editor's summary
Breaking down plastic into its original building blocks is an ideal recycling strategy in principle. Unfortunately, in practice, this approach isn't possible for the two most common plastics currently in use, polyethylene and polypropylene, because the reaction is too energetically unfavorable. Very recently, several groups of researchers showed that introducing fresh ethylene with the right catalyst can transform polyolefins into propylene, but the precious metals used for the catalysis are prohibitively expensive. Conk *et al*. now report that the process works using a more Earth-abundant combination of tungsten oxide and sodium. —Jake S. Yeston

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